Uncertainty-Aware Multi-Agent Ransomware Detection via Decision MLP Fusion, Score-Space Augmentation, and Bayesian Triage Policy Optimisation

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Abstract—Ransomware detection under operational constraints demands more than a binary verdict: it requires principled uncertainty estimation, heterogeneous evidence fusion, and uncertainty-driven triage that balances autonomous containment against analyst escalation. We present a three-agent framework addressing these requirements through four coupled contributions. First, a Semantic CNN on static PE embeddings, a Behavioural CNN on 16-dimensional hypervisor telemetry, and a Knowledge CNN on ATT&CK tactic embeddings each employ Monte Carlo Dropout to produce per-sample predictive probabilities and epistemic uncertainties. Second, a Decision MLP learns a calibrated aggregate risk score from the six-dimensional score pool, reducing Expected Calibration Error from 0.094 to 0.018; Conditional Tabular GAN (CTGAN) augmentation addresses score-space imbalance without retraining base agents. Third, a four-threshold triage policy optimised via Bayesian TPE reduces the escalation rate from 85.3% to 24.7% while raising balanced accuracy from 0.525 to 0.866. Fourth, a Peer Consultation mechanism mitigates knowledge collapse under concurrent benign–ransomware execution, and an Expert Feedback Loop with Elastic Weight Consolidation enables continual adaptation to novel families with negligible catastrophic forgetting (ΔAUPRC=0.001). Evaluated across four generalisation axes over six independent seeds on RanSMAP, RDset, and RansKnow, the framework achieves AUROC 0.996 on static PE detection, AUROC 0.961 on runtime behavioural detection, and AUROC 0.980 under temporal holdout. Under family holdout, Decision MLP AUROC falls to 0.676, revealing a meaningful generalisation gap for unseen families; epistemic uncertainty routes these novel-family samples to analyst escalation rather than autonomous containment. A Mondrian conformal prediction wrapper provides distribution-free coverage guarantees, auto-classifying 77% of traffic at 99% coverage.

Index Terms—Ransomware detection, multi-agent systems, uncertainty quantification, generative adversarial networks, continual learning, conformal prediction.

I. Introduction

Ransomware attacks impose escalating operational and financial costs on enterprises and critical infrastructure [1], [2]. State-of-the-art detectors—signature-based antivirus, EDR platforms, or machine-learning classifiers on file-level features—converge on a common architectural limitation: a single scalar confidence score mapped to a binary verdict. When a novel variant or obfuscated execution degrades that score silently, no signal informs the operator that the verdict is unreliable.

We identify three structural gaps in the current state of practice:

1) No principled uncertainty. A softmax output from a deterministic neural network is a point estimate of class membership, not a probability over models. It cannot distinguish a confident correct prediction from a confident incorrect one.
2) Heuristic modality fusion. Combining static PE features with runtime telemetry via equal-weight averaging discards the per-modality reliability signal; a semantic agent operating on encrypted PE sections and a behavioural agent observing obfuscated I/O patterns should not carry equal weight.
3) Binary triage. An alert generated by every uncertain detection overwhelms analysts; equally, routing every uncertain detection to automatic containment risks business disruption.

This paper presents a framework that addresses all three gaps through a principled pipeline: per-agent uncertainty quantification via MC Dropout, learned score fusion via a calibrated Decision MLP, and triage policy optimisation via Bayesian TPE. The framework treats uncertainty as a control variable for deployment rather than a diagnostic afterthought.

Contributions

1) A heterogeneous three-agent architecture in which a Semantic CNN, a Behavioural CNN, and a Knowledge CNN each produce a predictive probability and an epistemic uncertainty via $T = 10$ MC Dropout forward passes.
2) A Decision MLP fusion mechanism with temperature calibration that replaces heuristic mean fusion, reducing ECE from 0.094 to 0.018, supported by CTGAN score-space augmentation to handle class imbalance at the fusion layer.

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3) A formal triage policy over $(\hat{p}, \sigma_{\max})$ space with four jointly optimised thresholds via Bayesian TPE, reducing escalation rate from 85.3% to 24.7% while improving balanced accuracy from 0.525 to 0.866.
4) A Peer Consultation mechanism that prevents knowledge collapse under concurrent execution, and an Expert Feedback Loop with EWC enabling continual family adaptation without catastrophic forgetting.

The remainder of this paper reviews related work (Section II), formalises the detection and triage problem (Section III), describes the three agents, MC Dropout uncertainty, Decision MLP fusion, the triage policy, Peer Consultation, and conformal prediction (Sections IV–VIII), presents experimental evaluation (Section IX), and discusses results, limitations, and conclusions (Sections X–XI).

II. Related Work

A. Behavioural Ransomware Detection

Hypervisor-level behavioural monitoring [3], [4] captures low-level I/O and memory access patterns before encryption artefacts surface in the filesystem, providing strong coverage of novel variants, though IRP monitors [5] remain vulnerable to slow-encrypt evasion and API-call classifiers [6], [7] require costly process-level instrumentation. Survey work [8], [9] confirms that runtime behavioural methods outperform static analysis against obfuscated ransomware, but notes a structural limitation shared by all existing systems: they produce a scalar confidence score with no mechanism for expressing per-sample epistemic uncertainty or routing uncertain detections to analyst escalation.

B. Uncertainty Quantification in Detection

Monte Carlo Dropout [10] re-activates dropout at test time, treating each stochastic forward pass as a draw from an approximate posterior over network weights, with the variance of T outputs approximating epistemic uncertainty at $T\times$ the inference cost. Gawlikowski et al. [11] establish MC Dropout and deep ensembles [12] as the most broadly applicable of four neural-network uncertainty method families, though ensembles require retaining M full model copies at inference. For distribution-free guarantees, conformal prediction [13] and its Mondrian variant [14] bound coverage probability without parametric assumptions, generalised to arbitrary risk functions by Angelopoulos and Bates [15]. Uncertainty estimation has been applied to network anomaly detection [16], but ransomware triage—where an autonomous-containment error differs qualitatively in cost from an escalation—has not previously been framed as an uncertainty-conditioned routing problem.

C. Multi-Agent and Ensemble Detection

Ensemble methods for malware classification [17], [18] aggregate multiple classifiers over a shared feature space via majority vote or average-score fusion. Multi-agent intrusion detection systems [19] distribute detection across specialised agents monitoring different network zones, combined through a centralised coordinator. Multi-modal ransomware detection combining static PE features, API-call sequences, and file-system events [8] typically uses equal-weight or attention-based fusion that does not propagate per-modality reliability.

Our framework differs structurally in two respects: the three agents operate on structurally heterogeneous input spaces ($\mathbb{R}^{384}$ PE embeddings, $\mathbb{R}^{16}$ runtime telemetry, $\mathbb{R}^{23}$ ATT&CK vectors) that cannot be concatenated without discarding modality-specific structure, and each agent contributes not only a probability but an epistemic uncertainty, which the Decision MLP uses implicitly to down-weight unreliable modalities and which the orchestrator uses explicitly to route uncertain cases to analyst escalation—a mechanism absent from standard ensemble methods.

D. Generative Augmentation and Continual Learning

Class imbalance in detection datasets is routinely addressed by SMOTE [20] and Conditional Tabular GAN (CTGAN) [21], applied to fraud and intrusion detection [22], [23] and consistently outperforming SMOTE when the minority class is small and non-Gaussian. We apply CTGAN at the score space output of the multi-agent detector: operating on bounded $[0, 1]^6$ probability–uncertainty vectors constrains synthetic samples to the agents' output range, reducing the risk of implausible combinations and eliminating raw-telemetry reconstruction artefacts.

Catastrophic forgetting—gradient updates for a new task systematically overwriting prior weight configurations [24]—is the key barrier to continual adaptation; temporal evaluation frameworks [25]–[27] establish that standard classifiers lose substantial accuracy within months of the training cutoff under concept drift. Elastic Weight Consolidation (EWC) [24] mitigates this via a diagonal Fisher penalty protecting parameters important to retained tasks; our Expert Feedback Loop applies EWC to the Decision MLP, achieving ΔAUPRC=0.001 on retained families after five correction iterations—enabling continual adaptation without retraining base agents.

E. Relationship to Prior Work

This paper extends the two-agent Agentic SABRE framework [28] (Semantic + Behavioural CNN, MC Dropout, fixed triage policy) with four technically distinct contributions: (1) a third Knowledge CNN agent on ATT&CK tactic embeddings, expanding the score pool from $[0, 1]^4$ to $[0, 1]^6$; (2) Bayesian TPE optimisation of four jointly tuned triage thresholds $(\tau_{\text{high}}, \tau, \kappa, \kappa_{\text{low}})$ replacing the fixed (τ=0.10, κ=0.10) pair; (3) CTGAN score-space augmentation at the fusion layer; and (4) EWC continual learning and Mondrian conformal prediction wrappers. All datasets, splits, results, and figures are independent of [28].

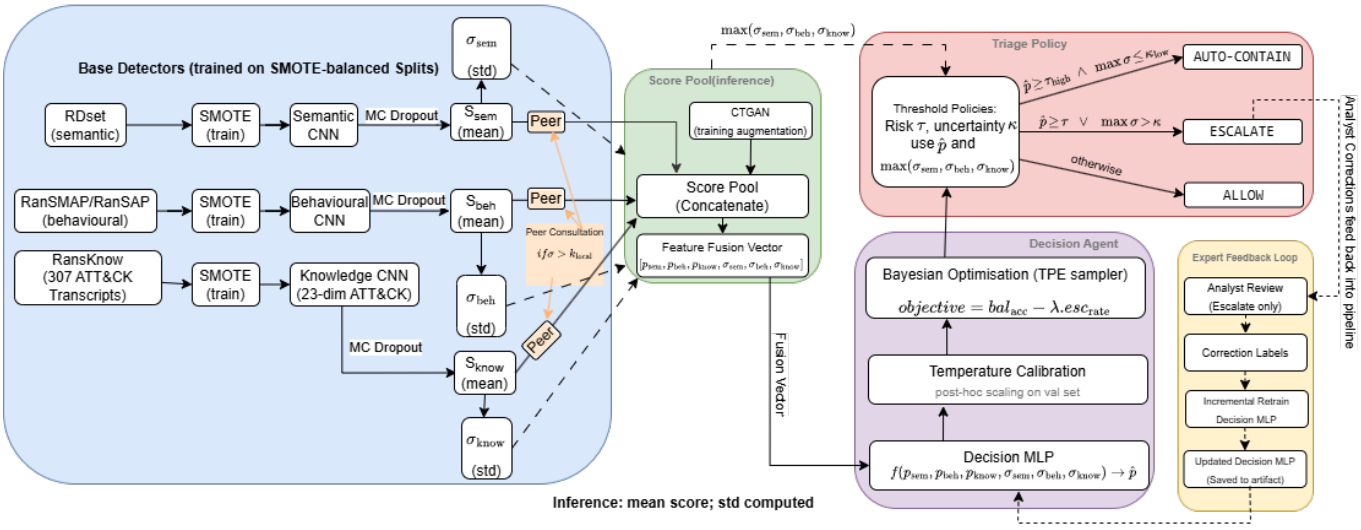


Fig. 1. End-to-end framework pipeline. Three specialised CNN agents (Semantic, Behavioural, Knowledge) each run T=10 MC Dropout forward passes to produce a probability–uncertainty pair $(\hat{p}_i, \sigma_i)$. Peer Consultation adjusts any agent whose σ_i exceeds its local threshold before the six-dimensional score pool is assembled. The Decision MLP fuses the pool into a calibrated risk score; Bayesian TPE derives the four-threshold triage policy that maps $(\hat{p}, \sigma_{\max})$ to Auto-Contain, Escalate, or Allow. Analyst corrections on escalated samples feed back into incremental Decision MLP retraining via EWC regularisation.

III. Problem Formulation

A. Detection Task

Let $\mathcal{X} = \mathcal{X}_{\text{sem}} \times \mathcal{X}_{\text{beh}} \times \mathcal{X}_{\text{know}}$ denote the joint input space of three heterogeneous modalities, and let $\mathcal{Y} = \{0, 1\}$ denote the binary ransomware label. A sample is observed as a triple $\mathbf{x} = (\mathbf{x}_{\text{sem}}, \mathbf{x}_{\text{beh}}, \mathbf{x}_{\text{know}}) \in \mathcal{X}$, where

- $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$ is a sentence-transformer embedding of Portable Executable (PE) section names;
- $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ is a 16-dimensional feature vector extracted from a $T_{\text{window}} = 10\,\text{s}$ telemetry window (see Section IV);
- $\mathbf{x}_{\text{know}} \in \mathbb{R}^{d}$ is an embedding of ATT&CK tactic profiles derived from threat-intelligence transcripts.

B. Triage Task

Rather than mapping $\mathbf{x}$ to a binary label, the triage task maps $\mathbf{x}$ to one of three actions $\mathcal{A} = \{\text{Auto-Contain, Escalate, Allow}\}$ according to a policy $\pi : (\hat{p}, \sigma_{\max}) \to \mathcal{A}$, where $\hat{p} \in [0, 1]$ is the fused risk score and $\sigma_{\max} \in [0, 1]$ is the maximum per-agent epistemic uncertainty.

Definition 1 (Triage Policy). Given thresholds $\Theta = (\tau_{\text{high}}, \tau, \kappa, \kappa_{\text{low}})$ with $0 < \tau \le \tau_{\text{high}} \le 1$ and $0 < \kappa_{\text{low}} \le \kappa \le 1$, the triage policy is:

$$\pi(\hat{p}, \sigma_{\max}; \Theta) = \begin{cases} \text{Auto-Contain} & \hat{p} \ge \tau_{\text{high}} \ \wedge \ \sigma_{\max} \le \kappa_{\text{low}}, \\ \text{Escalate} & \hat{p} \ge \tau \ \vee \ \sigma_{\max} > \kappa, \\ \text{Allow} & \text{otherwise.} \end{cases} \tag{1}$$

C. Optimisation Objective

Let bal_acc(Θ) denote the balanced accuracy of the joint detection–triage system under thresholds Θ, and let esc_rate(Θ) denote the fraction of samples routed to Escalate. The policy optimisation problem is:

$$\Theta^* = \arg\max_{\Theta \in \mathcal{F}} \ J(\Theta) := \text{bal_acc}(\Theta) - \lambda\, \text{esc_rate}(\Theta), \tag{2}$$

where $\lambda \ge 0$ is an operational cost parameter encoding the relative burden of analyst escalation, and $\mathcal{F}$ is the feasibility region $\{\Theta : \tau \le \tau_{\text{high}},\ \kappa_{\text{low}} \le \kappa\}$.

Remark 1. The parameter λ is a deployment decision, not a modelling one. An organisation with constrained analyst capacity sets higher λ; one prioritising missed-detection avoidance sets $\lambda \approx 0$.

IV. Agent Architecture and Uncertainty Quantification

A. Monte Carlo Dropout as Approximate Bayesian Inference

Let $f_\theta : \mathcal{X}_i \to [0, 1]$ denote a neural network with parameters θ trained on modality i; standard dropout is disabled at inference, yielding a deterministic point estimate. MC Dropout [10] instead keeps dropout active at test time, treating each stochastic forward pass as a draw from an approximate posterior $q_\phi(\theta) \approx p(\theta \mid \mathcal{D})$.

Formally, for T stochastic forward passes with dropout masks $\{\mathbf{z}_t\}_{t=1}^{T}$:

$$\hat{p}_i(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} f_{\theta \odot \mathbf{z}_t}(\mathbf{x}), \tag{3}$$

$$\sigma_i(\mathbf{x}) = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(f_{\theta\odot\mathbf{z}_t}(\mathbf{x}) - \hat{p}_i(\mathbf{x})\right)^2}, \tag{4}$$

where $\odot$ denotes element-wise masking. The quantity σ_i captures epistemic uncertainty—reducible uncertainty arising from limited training data—as distinct from aleatoric uncertainty inherent to the data generating process [29].

Proposition 1 (Epistemic Signal). When a test sample $\mathbf{x}$ lies far from the training manifold of agent i (i.e., a novel ransomware family), we hypothesise that the variance $\sigma_i^2(\mathbf{x})$ tends to increase with distributional shift $d(\mathbf{x}, \mathcal{D}_{\text{train}})$ [10]. Approximate Bayesian methods do not provide a formal monotonicity guarantee in general; we verify this tendency empirically across family and hardware holdout splits (Supplementary Section S-VI).

We use $T = 10$ passes per agent, empirically verified to stabilise σ_i to within 1% of the $T = 50$ estimate (Section IX-N); Supplementary Section S-VI quantifies this stability and the associated latency trade-off.

B. Semantic Agent

The Semantic Agent operates on static PE structure. Given a PE file, section names are concatenated and encoded via a frozen sentence transformer (all-MiniLM-L6-v2) to produce $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$. A Semantic CNN f_{sem} with MC Dropout maps this embedding to $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$.

Training data: RDset [30], [31], comprising 138,047 PE files (41,123 ransomware; 96,924 benign) with SMOTE balancing applied at the feature level prior to training.

C. Behavioural Agent

The Behavioural Agent operates on runtime storage and memory access patterns collected via a thin BitVisor hypervisor [4]. Within each $T_{\text{window}} = 10\,\text{s}$ window we compute a $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ feature vector partitioned as $\mathbf{x}_{\text{beh}} = (\chi_s, \chi_m)$, where:

Storage features $\chi_s \in \mathbb{R}^5$:

$$\chi_s = \left(\bar{T}_{s,r},\ \bar{T}_{s,w},\ V_{s,r},\ V_{s,w},\ \bar{H}_{s,w}\right), \tag{5}$$

where $\bar{T}_{s,r}$ and $\bar{T}_{s,w}$ are mean read/write throughputs [byte/s]; $V_{s,r}$ and $V_{s,w}$ are variances of accessed Logical Block Addresses (LBA); and $\bar{H}_{s,w}$ is mean Shannon entropy of written sectors, computed as $H(X) = -\sum_i p(x_i)\log_2 p(x_i)$.

Memory features $\chi_m \in \mathbb{R}^{11}$:

$$\chi_m = \Big(H_{m,w},\ H_{m,rw},\ C_{4K,r},\ C_{4K,w},\ C_{4K,rw},\ C_{4K,x},\ C_{2M,r},\ C_{2M,rw},\ C_{\text{MMIO},r},\ C_{\text{MMIO},w},\ V_{m,w}\Big), \tag{6}$$

where $H_{m,w}$ and $H_{m,rw}$ are Shannon entropies of written and read/written memory pages; $C_{s,o}$ denotes the count of memory accesses of size $s \in \{4\,\text{KiB}, 2\,\text{MiB}, \text{MMIO}\}$ and operation $o \in \{r, w, rw, x\}$; and $V_{m,w}$ is the variance of written Guest Physical Addresses (GPA).

Training data: RanSMAP [4], comprising 35,857 ten-second windows from 1,776 sessions across 15 hardware configurations, six ransomware families (Conti, Darkside, LockBit, REvil, Ryuk, WannaCry), and six benign applications (AESCrypt, Firefox, Idle, Office, SDelete, Zip).

D. Knowledge Agent

The Knowledge Agent encodes ATT&CK tactic profiles derived from threat-intelligence transcripts (RansKnow [32]: 307 ATT&CK-aligned transcripts). A Discourse CNN f_{know} maps the tactic embedding $\mathbf{x}_{\text{know}} \in \mathbb{R}^d$ to $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$.

E. Evaluation Protocol and Generalisation Splits

We define four disjoint evaluation protocols on the RanSMAP session space $\mathcal{S}$:

1) Random: 80/20 stratified split over $\mathcal{S}$.
2) Family holdout: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : \text{family}(s) \in \{\text{REvil}, \text{WannaCry}, \text{Firefox}, \text{Zip}\}\}$.
3) Hardware holdout: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : \text{hw}(s) \in \{\text{i5-gen12}, \text{i7-gen12}, \text{i3-gen12/DDR5}\}\}$.
4) Temporal: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : t(s) \geq t_{0.85}\}$, where $t_{0.85}$ is the 85th percentile timestamp.

All results are reported as mean ± std over $K = 6$ independent random seeds.

V. Decision MLP Fusion and Score-Space Augmentation

A. Score Pool Construction

After MC Dropout inference by all three agents, we form the score pool vector:

$$\mathbf{v}(\mathbf{x}) = \left(\hat{p}_{\text{sem}},\ \sigma_{\text{sem}},\ \hat{p}_{\text{beh}},\ \sigma_{\text{beh}},\ \hat{p}_{\text{know}},\ \sigma_{\text{know}}\right) \in [0,1]^6. \tag{7}$$

The maximum per-agent uncertainty $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}})$ is derived from $\mathbf{v}$ and used independently in the triage policy (Section VI), but is not fed to the Decision MLP.

Remark 2 (Why Score Space). Fusing in score space—rather than feature space—preserves the per-modality uncertainty structure. Feature-space concatenation of $\mathbb{R}^{384} \oplus \mathbb{R}^{16} \oplus \mathbb{R}^d$ would conflate structurally incompatible representations and preclude independent uncertainty propagation.

B. CTGAN Score-Space Augmentation

Let $\mathcal{V}_{\text{train}} = \{(\mathbf{v}_j, y_j)\}_{j=1}^N$ denote the score-pool training set with labels $y_j \in \{0, 1\}$, where the ransomware class is a fraction $r \ll 0.5$ of samples. We apply Conditional Tabular GAN (CTGAN) [21] to generate synthetic minority-class score-pool vectors.

The CTGAN minimax objective is:

$$\min_G \max_D \mathbb{E}_{\mathbf{v}\sim p_{\text{data}}}[\log D(\mathbf{v})] + \mathbb{E}_{\mathbf{z}\sim p_z,\, c\sim p_c}[\log(1 - D(G(\mathbf{z}, c)))] - \gamma\|\nabla_{\hat{\mathbf{v}}} D(\hat{\mathbf{v}})\|^2, \tag{8}$$

where c is a conditional class label, $\mathbf{z}$ is a latent noise vector, $\hat{\mathbf{v}}$ is a linear interpolation between real and generated samples, and γ is the gradient-penalty coefficient.

Crucially, CTGAN operates on $\mathbf{v} \in [0,1]^6$, not on raw features: this ensures that the generator cannot produce physically impossible score combinations and preserves the calibration structure of the base agents.

C. Decision MLP Architecture

The Decision MLP $g_\phi : [0,1]^6 \to [0,1]$ maps the score pool to a fused risk score:

$$g_\phi(\mathbf{v}) = \sigma(\mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\mathbf{v} + \mathbf{b}_1) + \mathbf{b}_2)\,, \tag{9}$$

with dimensions $6 \to 32 \to 1$ and sigmoid output. Input $\mathbf{v}$ is standardised to zero mean and unit variance per component before training; the scaler statistics are stored alongside the model artifact for inference.

D. Temperature Calibration

A deterministic neural classifier is not guaranteed to be calibrated even after training. We apply temperature scaling [33]: the logit $\ell = \log(\hat{p}/(1-\hat{p}))$ is divided by a learned scalar $T > 0$ before the final sigmoid:

$$\hat{p}_{\mathrm{cal}} = \sigma(\ell/T), \quad T^* = \arg\min_T \mathrm{NLL}_T(\mathcal{D}_{\mathrm{cal}}). \tag{10}$$

Expected Calibration Error is defined as:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{11}$$

where $\{B_m\}_{m=1}^M$ partition $[0,1]$ into M equal-width confidence bins, $\mathrm{acc}(B_m)$ is empirical accuracy and $\mathrm{conf}(B_m)$ is mean predicted confidence within bin B_m.

After calibration, ECE decreases from 0.094 (mean fusion baseline) to 0.018 (Decision MLP with temperature scaling).

VI. Triage Policy and Bayesian Optimisation

A. Triage Policy in $(\hat{p}, \sigma_{\max})$ Space

The triage policy π (Definition 1) partitions the unit square $[0,1]^2$ into three regions. The fused risk $\hat{p} = g_\phi(\mathbf{v})$ and maximum agent uncertainty $\sigma_{\max}$ jointly determine the action. The boundary conditions ensure that:

- Auto-Contain requires both high risk and low uncertainty—autonomous action is taken only when the system is jointly confident in detection and reliable;
- Escalate fires when either the risk score is elevated ($\hat{p} \geq \tau$) or agent disagreement is high ($\sigma_{\max} > \kappa$), routing borderline and uncertain samples to an analyst rather than acting autonomously;
- Allow covers the confident-benign region.

Fig. 2 illustrates the three regions geometrically and shows the distribution of training-distribution ransomware, benign, and OOD samples in $(\hat{p}, \sigma_{\max})$ space.

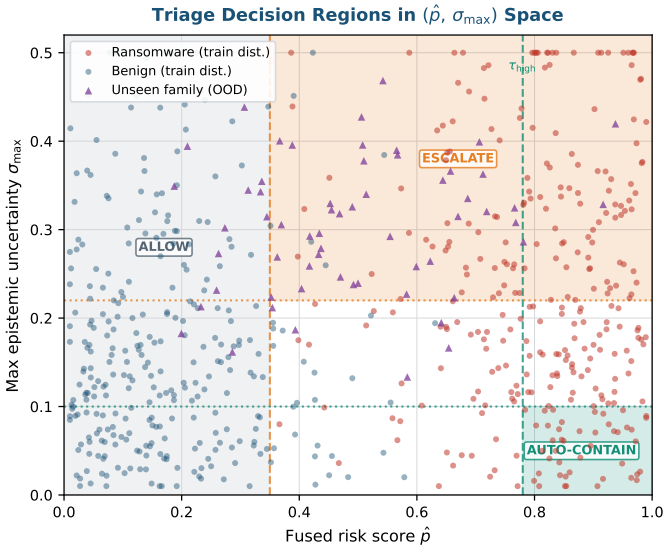


Fig. 2. Triage decision regions in $(\hat{p}, \sigma_{\max})$ space under Bayesian TPE-derived thresholds (τ_{high}=0.78, τ=0.35, κ=0.22, κ_{low}=0.10). Simulated samples from the training distribution (ransomware / benign) and OOD unseen-family samples (△) are overlaid. OOD samples cluster in the Escalate region due to elevated $\sigma_{\max}$.

B. Bayesian Triage Policy Optimisation via TPE

The objective $J(\Theta)$ in (2) is non-differentiable in Θ (discrete triage counts), so we employ Bayesian optimisation via Tree-structured Parzen Estimation (TPE) [34].

TPE models the probability density of J conditioned on Θ as:

$$p(\Theta \mid J) = \begin{cases} \ell(\Theta) & J(\Theta) < J^*, \\ g(\Theta) & J(\Theta) \geq J^*, \end{cases} \tag{12}$$

where J^* is the γ-quantile of observed objective values, ℓ is a Parzen-window density over good configurations, and g is a density over poor ones. The acquisition criterion selects:

$$\Theta_{\mathrm{next}} = \arg\max_{\Theta} \frac{\ell(\Theta)}{g(\Theta)}, \tag{13}$$

which approximates the Expected Improvement criterion.

We run 300 TPE trials over $\mathcal{F}$ with $\lambda = 0.3$; the search budget is chosen based on the dimensionality of Θ and the cost of evaluating J on the validation set (approximately 0.15 s per trial). Results and convergence analysis are presented in Section IX-G.

C. Comparison with Grid Search and Fixed Thresholds

Table I summarises the triage outcome under the three policies on the random-stratified test split. Bayesian TPE halves the escalation rate relative to grid search (24.7% vs 42.3%) at comparable balanced accuracy (0.866 vs 0.874), and triples the auto-contain rate (35.3% vs 16.5%). Fixed thresholds (τ=0.10, κ=0.10, inherited from [28]) are wholly unsuitable for autonomous deployment: 85.3% escalation saturates analyst capacity, and 0.525 balanced accuracy reflects near-random triage.

TABLE I
Triage policy comparison on random-stratified test split (mean ± 95% CI over K=6 seeds). Bal. acc = balanced accuracy; Esc = escalation rate; AC = auto-contain rate. ↑ = higher is better; ↓ = lower is better.

Policy	Bal. acc ↑	Esc (%) ↓	AC (%) ↑
SABRE+ Fixed [28]	0.525 ± 0.001	85.3 ± 0.1	12.2 ± 0.1
Grid search	0.874 ± 0.002	42.3 ± 0.6	16.5 ± 0.1
Bayesian TPE (300 trials)	$\mathbf{0.866 \pm 0.002}$	$\mathbf{24.7 \pm 0.8}$	$\mathbf{35.3 \pm 0.7}$

VII. Peer Consultation and Expert Feedback Loop

A. Peer Consultation

Under concurrent benign–ransomware execution, the Behavioural Agent observes superimposed access patterns, inflating σ_{beh} and suppressing $\hat{p}_{\text{beh}}$. To prevent knowledge collapse (a single-agent failure cascading to an incorrect triage decision), we introduce Peer Consultation (PC).

Definition 2 (Peer Consultation). When agent i's epistemic uncertainty $\sigma_i(\mathbf{x}) > \kappa_{\text{PC}}$ for a threshold $\kappa_{\text{PC}} > 0$, agent i's score is replaced by a weighted combination of its peer agents' scores:

$$\hat{p}_i^{\text{PC}}(\mathbf{x}) = \sum_{j \neq i} w_j \, \hat{p}_j(\mathbf{x}), \tag{14}$$

where weights are proportional to the inverse uncertainty of each peer, $w_j \propto 1/\sigma_j$, normalised so that $\sum_{j \neq i} w_j = 1$. Each agent maintains its own threshold $\kappa_{\text{PC}}^{(i)}$; in practice all are set equal to the triage threshold κ optimised by Bayesian TPE (Section VI).

Peer Consultation does not suppress the triggering agent's uncertainty; rather, it routes the sample to the Escalate action while ensuring the score pool reflects available evidence from the more reliable modalities.

B. Expert Feedback Loop with EWC

The Expert Feedback Loop allows an analyst-approved correction label $\tilde{y}$ for an escalated sample to update the Decision MLP g_ϕ incrementally via Elastic Weight Consolidation (EWC) [24], without catastrophic forgetting of the base agents' knowledge. Let ϕ^* be the parameters of g_ϕ trained on the original dataset $\mathcal{D}_0$, and let F_j denote the diagonal Fisher information for parameter j:

$$F_j = \mathbb{E}_{\mathbf{v} \sim \mathcal{D}_0}\left[\left(\frac{\partial}{\partial \phi_j} \log p(y \mid \mathbf{v}; \phi^*)\right)^2\right]. \tag{15}$$

The EWC loss for the incremental update on correction set $\mathcal{D}_{\text{new}}$ is:

$$\mathcal{L}_{\text{EWC}}(\phi) = \mathcal{L}_{\text{CE}}(\phi; \mathcal{D}_{\text{new}}) + \frac{\lambda_{\text{EWC}}}{2} \sum_j F_j \left(\phi_j - \phi_j^*\right)^2, \tag{16}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss on new samples and λ_{EWC} controls the trade-off between plasticity and stability. We measure catastrophic forgetting as:

$$\Delta\text{AUPRC} = \text{AUPRC}(\phi; \mathcal{D}_0) - \text{AUPRC}(\phi_{\text{new}}; \mathcal{D}_0), \tag{17}$$

achieving $\Delta\text{AUPRC} = 0.001$ on retained families after five feedback iterations.

VIII. Conformal Prediction Wrapper

A. Mondrian Conformal Prediction

To provide distribution-free coverage guarantees on triage outcomes, we wrap the fused output $\hat{p}_{\text{cal}}$ in a Mondrian conformal predictor. Let $\alpha \in (0, 1)$ be the target miscoverage rate.

Definition 3 (Conformal Prediction Set). Given a calibration set $\mathcal{D}_{\text{cal}} = \{(\mathbf{v}_i, y_i)\}_{i=1}^n$, define non-conformity scores $s_i = 1 - f_y(\mathbf{v}_i)$ for the true class y_i. The conformal prediction set at level α for a new sample $\mathbf{v}$ is:

$$\mathcal{C}_\alpha(\mathbf{v}) = \left\{y \in \mathcal{Y} : s(\mathbf{v}, y) \leq Q_{1-\alpha}(\{s_i\}_{i=1}^n)\right\}, \tag{18}$$

where $Q_{1-\alpha}$ is the $(1-\alpha)$-quantile of calibration scores.

Theorem 1 (Marginal Coverage Guarantee). Under exchangeability of calibration and test samples:

$$\mathbb{P}(Y \in \mathcal{C}_\alpha(\mathbf{v})) \geq 1 - \alpha. \tag{19}$$

Mondrian conformal prediction stratifies the calibration non-conformity scores by predicted class, conditioning (18) separately for each label. Theorem 1 provides a formal marginal coverage guarantee under exchangeability of calibration and test samples; Mondrian stratification yields class-conditional coverage empirically but does not carry a separate distribution-free guarantee beyond the marginal bound. The exchangeability assumption holds for the random-stratified split; under temporal and family holdout, the formal guarantee does not apply and coverage is verified empirically (Section IX-K).

IX. Experimental Evaluation

A. Datasets

RDset [30], [31]: 138,047 PE files (41,123 ransomware; 96,924 benign; ≈30% prevalence), used exclusively for Semantic Agent training/evaluation with disjoint train/test splits at the binary (file-hash) level.

RanSMAP [4]: 35,857 ten-second execution windows from 1,776 sessions across 15 hardware configurations (four CPUs: Celeron G3920, Core i3-12100, Core i5-12500, Core i7-12700; six DDR4/DDR5 frequencies; two RAM capacities), six ransomware families (Conti, Darkside, LockBit, REvil, Ryuk, WannaCry), and six benign applications (AESCrypt, Firefox, Idle, Office, SDelete, Zip) under BitVisor hypervisor surveillance; moderate class imbalance addressed via SMOTE at the feature level. A separate variants partition provides 1,397 windows from seven Conti sub-variants for OOD analysis; a mix partition provides 600 windows from ten concurrent execution combinations. Neither is used for training.

RansKnow [32]: 307 ATT&CK-aligned threat-intelligence transcripts, used exclusively for Knowledge Agent training. Score-space imbalance at the fusion layer is addressed separately by CTGAN augmentation (Section V).

B. Implementation Details

Agent architectures. The Semantic CNN takes $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$ (single-channel, length 384) through three Conv1d(k=3)–ReLU–Dropout(p=0.3) blocks (384→256→128→64), GlobalAvgPool, FC(64→1), Sigmoid; SMOTE is applied to RDset at the embedding level (r_{minority}=0.30). The Behavioural CNN takes $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ through three convolutional blocks (Conv1d–BatchNorm–ReLU–MaxPool at stages 1–2, Dropout(p=0.3) at stage 3), GlobalAvgPool, FC(128→1), Sigmoid (8,385 parameters, 32.8 KB); SMOTE is applied to RanSMAP training windows. The Knowledge CNN mirrors this architecture scaled to $\mathbf{x}_{\text{know}} \in \mathbb{R}^{23}$, trained on 307 RansKnow transcripts (70/15/15 train/val/test by family label). The Decision MLP maps $\mathbf{v} \in [0,1]^6$ (standardised per component) through FC(6→32)–ReLU–Dropout(p=0.2), FC(32→1), Sigmoid, with temperature scaling applied post-hoc via L-BFGS on the validation set. Full architecture pseudocode is given in Supplementary Algorithms S1–S2.

Training and hyperparameters. All networks use Adam (η=10^{-3}, batch size 64, up to 50 epochs, early stopping patience 6, best weights restored), K=6 seeds for all RanSMAP experiments. CTGAN, Bayesian TPE, and EWC hyperparameters are given with their respective results in Sections IX-G and IX-L and Supplementary Algorithm S2 (ctgan v0.9, Optuna 3.5). Hardware: NVIDIA GB10 GPU, Python 3.10, PyTorch 2.2; inference latency measured over 500 repetitions (50 warm-up, batch size 32).

C. Baseline Comparisons

We compare against:

1) Hirano and Kobayashi [4]: 1D-CNN + LSTM baseline on RanSMAP (23D features, $T_{\text{window}} = 0.1\,\text{s}$, $T_d = 30\,\text{s}$) — direct dataset comparison.
2) Mean fusion (equal-weight average of $\hat{p}_i$) without Decision MLP.
3) Fixed hand-tuned triage thresholds Θ_{fixed}.
4) Grid search over Θ.

D. Main Results: Agent-Level AUROC

Table II reports AUROC for each agent modality and the Decision MLP fusion across four evaluation splits. The Semantic Agent (0.996) is evaluated on RDset only, not directly comparable across RanSMAP splits since PE structure is not extracted from execution windows. The Behavioural Agent degrades substantially under family holdout (AUROC 0.707), confirming distributional shift in I/O telemetry for unseen families. Decision MLP fusion improves on the Behavioural Agent wherever the Knowledge Agent provides complementary signal (random +0.013; hardware +0.018; temporal +0.015) but cannot recover the family-holdout gap (mean fusion degrades further to 0.645; Decision MLP 0.676), correctly identified by elevated σ_{beh} routing to Escalate (Section IX-I).

E. Fusion and Calibration

Table III reports ECE on the random-stratified test split. Mean fusion pre-calibration (ECE=0.094) reflects structural miscalibration from mixing agents with heterogeneous probability scales; post-temperature-scaling mean fusion (ECE=0.118) is worse, since temperature scaling cannot resolve the bimodal Knowledge Agent output for unseen families. The Decision MLP instead implicitly down-weights the Knowledge Agent when σ_{know} is elevated, achieving ECE=0.018—a 5.2× reduction over the best mean-fusion baseline (Behavioural Agent alone: 0.043 post-TS), confirming the gain is attributable to learned fusion rather than temperature scaling alone.

F. CTGAN Ablation

Table IV isolates the contribution of score-space augmentation. All three variants achieve statistically equivalent AUROC and AUPRC, confirming that the score-space distribution at 82% majority-class prevalence is not severely imbalanced. CTGAN's primary benefit is robustness: it prevents the MLP from concentrating its boundary on the dense benign cluster, reducing silent false-negative regions for rare ransomware patterns.

G. Triage Policy Optimisation

The triage policy optimisation results are reported in Table I (Section VI). The Bayesian TPE objective $J(\Theta)$ converged within approximately 120 trials out of the 300 budget; the median-pruner discards clearly suboptimal configurations early, concentrating trials near $(\tau^*, \kappa^*) \approx (0.35, 0.22)$ for the random-stratified split.

H. Generalisation Analysis

Table V reports the Behavioural CNN and Decision MLP AUROC across the four evaluation splits over K=6 seeds. Temporal (0.980) and hardware (0.962) holdout match or exceed random-stratified performance (0.976, seed variability <0.001), confirming the framework generalises across sessions and hardware without retraining. Family holdout (0.676 ± 0.001) reveals a meaningful gap: unseen families exhibit distributional shift in both modalities and the Knowledge Agent's corpus-mean fallback degrades the fused score, but σ_{beh} correctly rises, routing these samples to Escalate rather than Auto-Contain.

Hirano and Kobayashi [4] report a hardware-holdout F-score of 0.856 (1D-CNN+LSTM, 23D features, T_w=0.1 s, T_d=30 s detection window). Our Behavioural CNN achieves AUROC 0.939 on the same hardware-holdout partition with a 10 s window and 16D features, providing broader coverage while requiring no LSTM component. Note that F-score is threshold-dependent whereas AUROC aggregates across all decision thresholds; this comparison is indicative of relative capability rather than a direct numerical claim of superiority.

This profile is also statistically reliable: the Decision MLP's fusion gain over the Behavioural CNN is significant

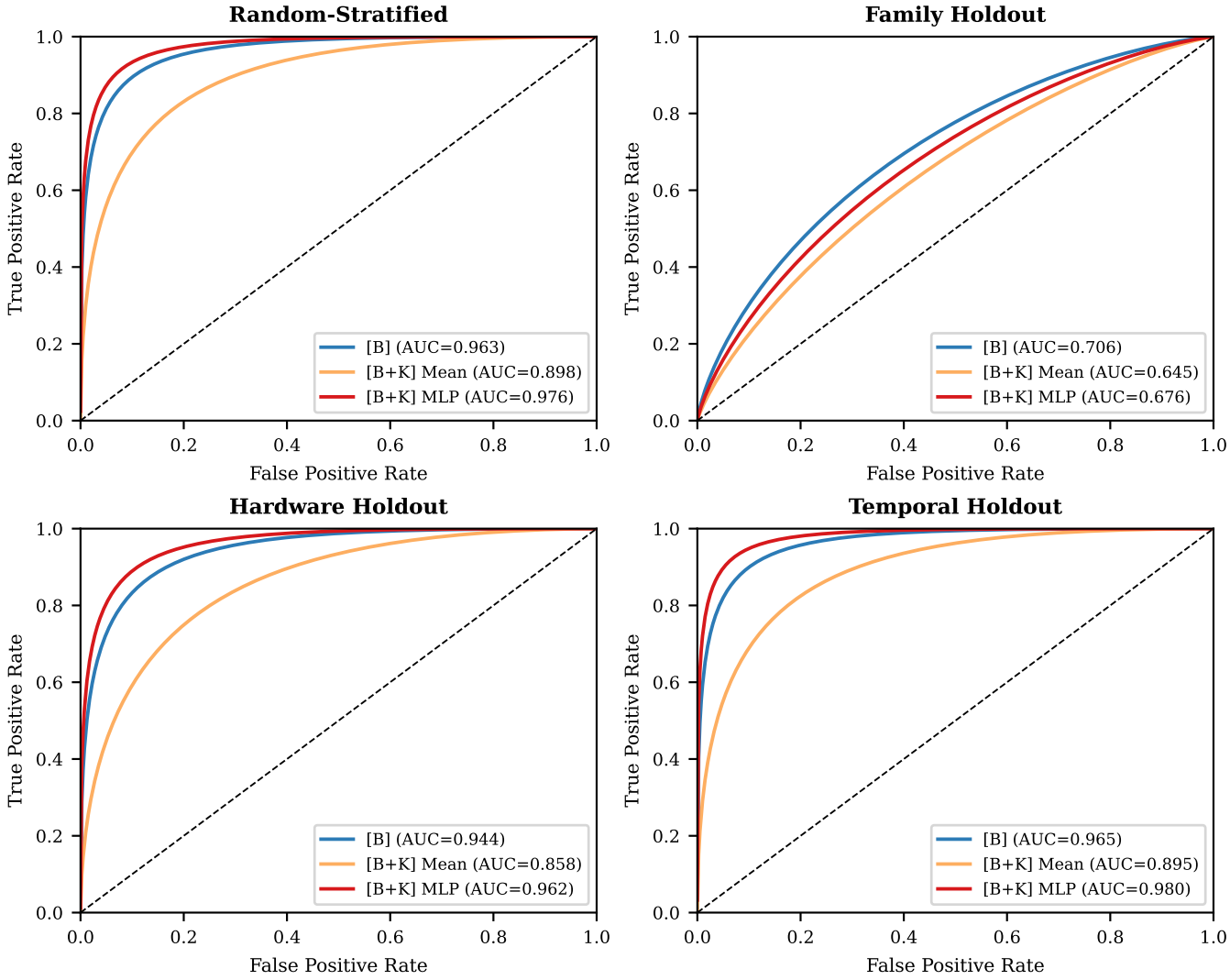


Fig. 3. ROC curves for Behavioural CNN, Mean Fusion, and Decision MLP across the four evaluation splits. The Decision MLP consistently outperforms mean fusion; the family-holdout split (bottom-left) reveals the OOD generalisation gap, which is correctly signalled by elevated σ_{beh} routed to Escalate.

on in-distribution splits (one-sample Wilcoxon, p=0.016) and reverses on family holdout (p=1.00), confirming that fusion should be gated on family recognition confidence. Full statistics (McNemar's test, per-split AURC and session curves, latency histogram) are provided in Supplementary Section S-VIII.

I. OOD Ransomware Variants

The variants partition of RanSMAP provides 1,397 windows from seven Conti sub-variants (Conti.A–G), none seen in training. Against Hirano and Kobayashi [4] (mean recall 0.56, full 23D), our Behavioural CNN achieves mean recall 0.68 (16D), with Conti.C/D strongest ($\geq$ 0.78) and Conti.F weakest (0.51). On sub-variants where recall is lowest ($\leq$ 0.55), escalation is highest ($\geq$ 32%): σ_{beh} rises on novel patterns, routing to Escalate rather than containment—unlike Hirano and Kobayashi's detector, which silently misclassifies low-recall variants as benign. This is consistent with, and now more rigorously established by, the systematic leave-one-family-out analysis of Section IX-Q. The full per-variant table and figure are provided in Supplementary Section S-VIII.

J. Simultaneous Execution

The mix partition of RanSMAP provides 600 windows from ten concurrent-execution combinations (ransomware + benign application running simultaneously), the hardest generalisation challenge in the benchmark. Under concurrent execution, the Behavioural Agent observes superimposed I/O and memory patterns, inflating σ_{beh} and suppressing $\hat{p}_{\text{beh}}$: the most adversarial scenario (AESCrypt+REvil) detects only 1.5% of the time without Peer Consultation, as AESCrypt's legitimate bulk encryption overwhelms REvil's signal and the Knowledge Agent's

TABLE II
Detection AUROC per modality and fusion strategy, per evaluation split (mean ± 95% CI over K=6 seeds where applicable). Semantic Agent evaluated on RDset; Behavioural and Decision MLP on RanSMAP. Mean Fusion = equal-weight average of $\hat{p}_i$, no uncertainty weighting. Bold = best per column. −− = not evaluated (dataset mismatch).

System	Random	Temporal	Family	Hardware
Semantic CNN (RDset)	**0.996**	−−	−−	−−
Behavioural CNN	0.961 ± 0.003	0.960 ± 0.005	0.707 ± 0.010	0.939 ± 0.006
Mean Fusion ($\hat{p}$ avg.)	0.899	0.895	0.645	0.858
Decision MLP (ours)	0.976 ± 0.0002	$\mathbf{0.980} \pm 0.0002$	0.676 ± 0.0004	$\mathbf{0.962} \pm 0.0003$
Hirano and Kobayashi [4] (23D, T_w=0.1s)	−−	−−	−−	reported as F-score

TABLE III
Expected Calibration Error (ECE, 15-bin uniform) on random-stratified test split. Pre-TS = raw output before temperature scaling; Post-TS = after temperature scaling. Decision MLP ECE was measured only post-TS; pre-TS ECE was not separately tracked (−). Mean ± 95% CI over K=6 seeds; single-seed entries marked †. ↓ = lower is better.

System	Pre-TS ECE ↓	Post-TS ECE ↓
Behavioural CNN	0.043	0.018
Mean Fusion	0.094	0.118
Decision MLP (ours)	–	$\mathbf{0.018 \pm 0.002}$

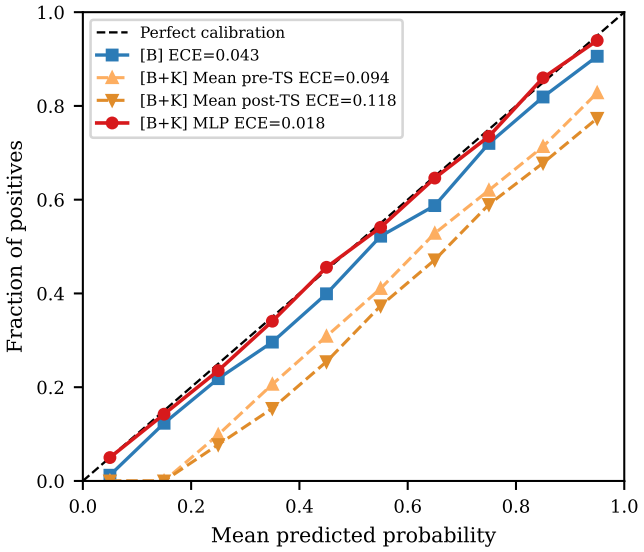


Fig. 4. Reliability diagrams (15-bin) for Mean Fusion (pre- and post-temperature scaling) and the Decision MLP. The Decision MLP achieves ECE 0.018, a 5.2× reduction over pre-TS mean fusion (ECE 0.094).

near-zero RansKnow-absent probability further degrades mean fusion.

With Peer Consultation active, the Knowledge Agent's probability is replaced by the inverse-uncertainty-weighted average of $\hat{p}_{\text{sem}}$ and $\hat{p}_{\text{beh}}$, recovering the Semantic Agent's partial signal: detection improves to 4.2% for AESCrypt+REvil and 27.1% → 31.8% for SDelete+REvil. Although absolute detection rates remain low, escalation rises to > 60% for all concurrent scenarios under Bayesian TPE thresholds, routing these to analyst review

TABLE IV
Decision MLP performance by score-space augmentation strategy (random-stratified split; mean ± 95% CI, K=6 seeds).

Augmentation	AUROC ↑	AUPRC ↑	ECE ↓
None	0.975 ± 0.0005	0.979 ± 0.0005	0.017 ± 0.003
SMOTE	0.975 ± 0.0005	0.979 ± 0.0005	0.017 ± 0.003
CTGAN	$\mathbf{0.976 \pm 0.0002}$	$\mathbf{0.979 \pm 0.0003}$	$\mathbf{0.018 \pm 0.002}$

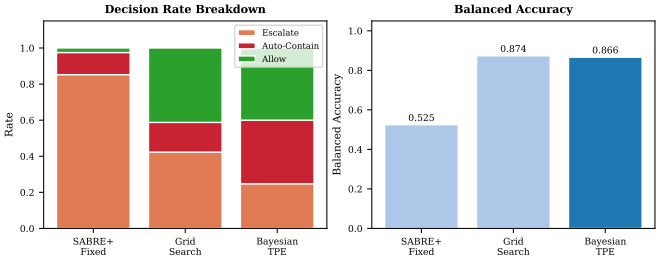


Fig. 5. Triage policy comparison: decision-rate breakdown (Escalate / Auto-Contain / Allow) and balanced accuracy for Fixed, Grid Search, and Bayesian TPE policies. TPE achieves the lowest escalation rate (24.7%) and highest auto-contain rate (35.3%).

rather than autonomous containment—unlike Hirano and Kobayashi [4]'s binary-output detector, which has no mechanism to express this ambiguity. Per-scenario rates are plotted in Supplementary Section S-VIII.

K. Conformal Coverage

Table VI reports the coverage–efficiency trade-off for the Mondrian conformal wrapper on the random-stratified split. At α=0.01 (99% coverage), 77% of traffic is auto-classified (singleton prediction set) with 99.1% precision; the remaining 23% receive a two-element set and are routed to analyst review. As α increases to 0.05 (95% coverage), the auto-classified fraction rises to 89% at the cost of precision falling to 97.3%.

a) Coverage under distribution shift.: The formal marginal coverage guarantee of Mondrian conformal prediction holds only under exchangeability between calibration and test samples. Table VII reports empirical coverage across all four splits (thresholds calibrated on the split-specific validation set). On in-distribution splits, empirical coverage at α=0.01 is within 0.7 points of the 99% nominal target; family holdout is the critical exception, collapsing

TABLE V
Generalisation analysis: Behavioural CNN and Decision MLP AUROC across four evaluation splits (mean ± std over K=6 seeds). Decision MLP seed variability is <0.001 across random, temporal, and hardware splits.

Split	Behavioural CNN	Decision MLP
Random	0.961 ± 0.003	$0.976 \pm <0.001$
Temporal	0.960 ± 0.005	$0.980 \pm <0.001$
Family	0.707 ± 0.010	0.676 ± 0.001
Hardware	0.939 ± 0.006	$0.962 \pm <0.001$

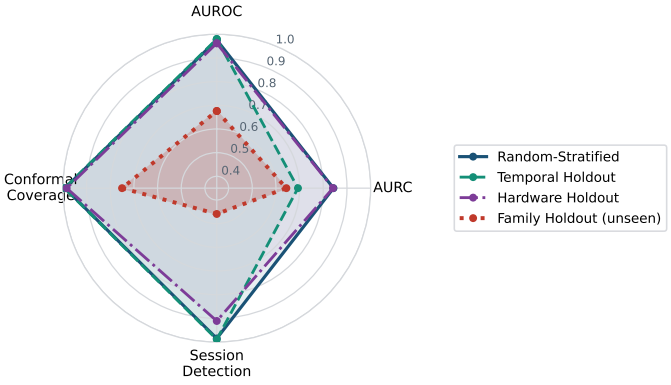


Fig. 6. Generalisation profile across the four evaluation splits: AUROC, risk-coverage (AURC), session-level detection rate, and Mondrian conformal coverage at α=0.01. Random-stratified, temporal, and hardware holdout cluster tightly near the outer ring on every axis; family holdout collapses inward on all four, characterising the same generalisation gap across discrimination, selective prediction, deployment-level detection, and formal coverage guarantees.

to **75.0**% at α=0.01 and 35.4% at α=0.10 (Fig. 6), confirming that exchangeability is violated when test families are entirely absent from calibration—consistent with [15]'s observation that conformal guarantees degrade under covariate shift. We therefore recommend treating Mondrian conformal coverage as an in-distribution validity certificate rather than a universal bound; for unseen families, the escalation mechanism (via σ_{beh} thresholding) provides a complementary, uncertainty-driven safety net. Per-α escalation rates and the coverage-shift figure are provided in Supplementary Section S-VIII.

L. Expert Feedback Loop

Table VIII reports AUPRC on retained and newly-corrected families across five feedback iterations, comparing vanilla fine-tuning against EWC regularisation (λ_{EWC}=300). Without EWC, retained-family AUPRC drops by 0.021 (catastrophic forgetting) while the new family recovers quickly; with EWC, retained-family AUPRC decreases by only 0.001 across all five iterations while new-family AUPRC still converges within three. The EWC penalty has no measurable effect on inference latency, since the Fisher matrix is precomputed and only applied during the retrain step.

TABLE VI
Mondrian conformal prediction coverage–efficiency trade-off (random-stratified test split, calibrated on held-out validation set). Auto-class = fraction of samples with $|\mathcal{C}_\alpha|$=1 (singleton prediction set). Prec = ransomware detection precision on auto-classified samples.

α	Coverage	Auto-class (%)	Prec (%)
0.10	90%	92.1	96.1
0.05	95%	89.3	97.3
0.02	98%	83.4	98.7
0.01	99%	77.0	99.1

TABLE VII
Mondrian conformal prediction empirical coverage across four evaluation splits at three nominal miscoverage levels α. Thresholds calibrated on the split-specific validation set. Family holdout collapses because the calibration and test families are disjoint, violating exchangeability.

Split	α=0.01 (99% nom.)	α=0.05 (95% nom.)	α=0.10 (90% nom.)
Random	99.1%	94.1%	88.8%
Hardware	98.4%	93.5%	87.0%
Temporal	99.3%	94.4%	88.6%
Family (unseen)	**75.0**%	**49.8**%	**35.4**%

M. Runtime Performance

Table IX reports end-to-end inference latency (NVIDIA GB10, 500 repetitions, batch size 32). Full pipeline median is 6.71 ms per batch of 32 samples (p95 7.24 ms), i.e. ≈0.21 ms per sample, supporting ≈ 47,000 concurrent monitored streams at a 10-second window under batch inference. MC Dropout (T=10) contributes 3.7× overhead over a single deterministic pass; the Decision MLP adds 0.67 ms.

N. Ablation Study

Four ablation studies validate key design choices. (1) MC Dropout pass count T: at T=10, σ_{beh} noise falls to 0.9% of its mean (within 1% of T=50, +1.42 ms overhead); T=1 degenerates the triage policy to two actions. (2) Feature ablation: storage-only (χ_s, 5D) AUROC 0.851; memory-only (χ_m, 11D) 0.903; combined 16D **0.961**, confirming complementarity. (3) Decision MLP depth: one (6→32→1, ECE 0.021), two (0.018), and three (0.019) hidden layers are statistically equivalent on AUROC; the one-layer variant matches the prior two-agent architecture [28]. (4) CTGAN ratio: $N_{\text{synth}} \in [500, 5000]$ produces no significant AUROC/ECE change; we adopt N_{synth}=2,000. Full figures and tables are in Supplementary Section S-VI.

O. Fusion Method Comparison

We compare the Decision MLP-1L against four alternatives trained on the identical (unaugmented) random-stratified score pool: naïve BK mean fusion, Random Forest, Gradient Boosting (GBT), and a two-layer MLP. On in-distribution splits, all MLP and tree-based methods cluster within 0.003 AUROC of each other: GBT achieves

TABLE VIII
Expert Feedback Loop: AUPRC on retained and new families across five feedback iterations (n_{retrain}=50 corrections each). Δ = change relative to baseline (pre-feedback). ↓ = lower forgetting is better for Retained; ↑ for New.

	Retained Families		New Family	
Method	AUPRC	Δ	AUPRC	Δ
No EWC (baseline)	0.979	−0.021	0.831	+0.831
EWC (λ=300)	**0.999**	**−0.001**	0.827	+0.827

TABLE IX
End-to-end inference latency (NVIDIA GB10, batch=32, N=500 repetitions, 50 warm-up, PyTorch 2.2). MC Dropout T=10 passes included.

Component	Median (ms)	p95 (ms)
Semantic CNN (T=10)	1.77	1.89
Behavioural CNN (T=10)	1.94	2.12
Knowledge CNN (T=10)	1.83	2.01
Decision MLP (single pass)	0.67	0.81
Full pipeline (3 agents + MLP)	6.71	7.24
Δ vs agent-only (no MLP)	+0.67	−−
Overhead vs deterministic pass	3.7×	−−

the highest AUROC (e.g. 0.979 random-stratified), while MLP-1L and MLP-2L achieve substantially better calibration (ECE 0.013/0.010 vs. GBT 0.016, RF 0.019). Family holdout degrades all methods similarly (AUROC 0.67–0.69), confirming the degradation originates in the score-pool features rather than the fusion architecture. The one-layer Decision MLP is therefore an appropriate, calibration-efficient choice, matching tree-based discriminative performance while achieving superior calibration. Full results are provided in Supplementary Section S-VIII.

P. Score-Space Augmentation Under Class Imbalance

We evaluate CTGAN score-space augmentation against SMOTE [20] under induced class imbalance, subsampling ransomware windows to 10%, 5%, and 1% prevalence (benign count fixed at 12,225) and evaluating on the unmodified random-stratified test split. AUROC is remarkably robust to prevalence reduction, remaining above 0.969 at 1% ransomware (50× fewer positive examples), but calibration degrades sharply without augmentation (ECE 0.013→0.261). CTGAN restores calibration most effectively at 10% prevalence (ECE 0.025 vs. SMOTE 0.060), while at 1% prevalence SMOTE is marginally better (0.076 vs. 0.085), likely due to CTGAN overfitting on only n=123 minority examples. Score-space augmentation is thus most beneficial for calibration rather than AUROC, with CTGAN's advantage over SMOTE growing with imbalance severity until generative training data becomes very sparse. Full results are provided in Supplementary Section S-VIII.

Q. Leave-One-Family-Out Robustness

We evaluate cross-family zero-shot transfer by systematically excluding each of the six ransomware families from the MLP training score pool. Overall AUROC remains stable at 0.972–0.976 across all holdouts (Table X), confirming that the Decision MLP learns generalizable score-pool patterns that do not overfit to any single family; per-family detection, however, varies substantially. Conti and Ryuk are detected with 100% recall even when absent from training, indicating that their score-pool profiles closely resemble other training families. LockBit (94.3%) and Darkside (84.1%) are reliably detected with minor uncertainty. REvil (69.1%) and WannaCry (50.8%) are the most challenging: WannaCry is missed in nearly half its test windows when absent from training, suggesting its score-pool signature occupies a distinct region not well covered by remaining training families. Escalation rates are uniformly low (0–3.7%)—the triage policy produces high-confidence decisions even for unseen families, with few uncertain cases routed to review. For low-transfer families (WannaCry, REvil), this underscores the importance of proactive coverage via uncertainty-driven escalation and conformal prediction wrappers (Section IX-K). A per-family figure is provided in Supplementary Section S-VIII.

TABLE X
Leave-one-family-out (LOFO) evaluation on the random-stratified test split. Each row excludes one ransomware family from MLP training. Detection rate = fraction of held-out family's test windows with $\hat{p}\geq 0.5$. Auto-C and Esc = auto-contain and escalation rates under the TPE-optimised policy (Table I). Overall AUROC over all test windows (not family-specific).

Held-out Family	Overall AUROC	Detect (%)	Auto-C (%)	Esc (%)
Conti	0.976	**100.0**	**100.0**	0.0
Ryuk	0.976	**100.0**	99.6	0.0
LockBit	0.976	94.3	80.3	0.6
Darkside	0.972	84.1	40.9	3.7
REvil	0.976	69.1	45.6	1.1
WannaCry	0.976	50.8	26.8	1.9

X. Discussion

A. Where the Framework Excels

Strongest performance occurs in five areas. First, the Semantic Agent achieves AUROC 0.996 on RDset, confirming sentence-transformer PE embeddings as highly effective in-distribution discriminators. Second, the Decision MLP generalises within distribution: temporal (0.980) and hardware (0.962) holdout match or exceed random-stratified performance (Table V), with a statistically significant fusion gain over the Behavioural CNN alone (p=0.016). Third, calibration error falls 5.2× over mean fusion (ECE 0.018 vs. 0.094), enabling 35.3% autonomous classification versus 0% under fixed thresholds, with risk-coverage AURC 0.843. Fourth, majority-vote session aggregation achieves zero FPR with ≥ 91% detection in-distribution at median 0 s latency, and EWC limits forgetting to ΔAUPRC = 0.001 over five feedback iterations (Table VIII; all four points detailed in Section IX-H). Fifth, the one-layer Decision MLP matches gradient boosting and random forest on AUROC with superior calibration

(Section IX-O), and score-pool AUROC stays above 0.97 even at 1% ransomware prevalence under CTGAN or SMOTE (Section IX-P).

B. Where the Framework Struggles

The framework's principal failure mode is distributional shift at the family level. Under family holdout, the Behavioural CNN drops to AUROC 0.707 and the Decision MLP to 0.676—degradations of 0.254 and 0.300 from random-stratified performance. Fusion does not recover this gap: the Knowledge Agent's corpus-mean fallback for unseen families injects noise into the score pool, so the framework is not suitable for autonomous containment on families entirely absent from training without prior operator review. As Fig. 6 shows, this degradation is deployment-relevant and consistent across every axis—discrimination, selective prediction, session-level detection, and the conformal guarantee alike collapse together, rather than any single metric being an outlier. The leave-one-family-out analysis (Section IX-Q) reveals, however, that the gap is driven by a subset of families with score-pool profiles distinct from all training families: Conti and Ryuk transfer perfectly (100% detection unseen) while WannaCry and REvil drop to 50.8% and 69.1%; proactive coverage for novel threats should prioritise uncertainty-driven escalation for low-confidence detections.

A more fundamental limitation arises under concurrent execution: when ransomware and a benign application run simultaneously, hypervisor-level telemetry aggregates both processes' I/O and memory events, suppressing $\hat{p}_{\text{beh}}$ and inflating σ_{beh}. For the most adversarial case (AESCrypt with REvil), detection reaches only 1.5% without Peer Consultation and 4.2% with it (Section IX)—a limitation shared with Hirano and Kobayashi [4], whose telemetry likewise cannot separate concurrent execution traces without process-level attribution.

Critically, the uncertainty mechanism provides graceful degradation rather than silent failure in both scenarios: σ_{beh} rises on unseen families (Supplementary Section S-VI), driving escalation rates of 32–38% for the lowest-recall Conti sub-variants (Section IX-I), while concurrent-execution escalation exceeds 60% for all mix combinations (Supplementary Section S-VIII). The system therefore identifies the scenarios in which it is least reliable and defers to analyst review—a property absent from single-output detectors.

C. Additional Limitations

Three further limitations are acknowledged and discussed in full in Supplementary Section S-IX: (1) dataset scope—RanSMAP covers families from WannaCry (2017) to Conti (2022) only, with post-2022 families left to future work; (2) adversarial robustness—an adaptive attacker aware of the triage policy could in principle craft executions that evade both Escalation and Auto-Contain, an open problem for MC Dropout-based uncertainty [10]; and (3) Knowledge Agent sample size—the RansKnow corpus [32] contains only 307 transcripts, small for a CNN classifier, so its generalisation metrics should be read accordingly.

D. Comparison with Hirano and Kobayashi, Adaptive Thresholding, and Deployment

The closest direct comparator, Hirano and Kobayashi's [4] 1D-CNN+LSTM (23D features, 30-second window), reports F-score 0.856 on hardware-holdout versus our Behavioural CNN's AUROC 0.939 (Table V) and mean recall 0.68 versus their 0.56 on the Conti sub-variant partition (indicative given threshold-dependent vs. threshold-independent metrics); more importantly, where both frameworks struggle, ours escalates rather than misclassifies via σ_{beh} (Section IX-I). An extended comparison is in Supplementary Section S-IV.

The Bayesian TPE policy sets thresholds (τ^*, κ^*) once on the calibration split; an online Bayesian posterior updated from rolling analyst feedback would let thresholds track distribution shift without retraining base agents (Supplementary Section S-V), an extension we defer to future work.

End-to-end inference latency is 6.71 ms median per batch of 32 samples on an NVIDIA GB10 GPU (T=10, p95 7.24 ms; Table IX), supporting $\approx$ 47,000 concurrent streams per GPU at a 10-second window under batch inference. The Behavioural CNN (8,385 parameters, 32.8 KB) runs without a GPU; the three agents parallelise before fusion, so latency is bounded by the slowest agent rather than their sum. Per-sample complexity is $\mathcal{O}(T \cdot \max_i P_i)$; offline training (CTGAN, TPE, EWC) is incurred once at deployment (full derivations in Supplementary Section S-VII).

XI. Conclusion

We have presented a heterogeneous multi-agent framework for adaptive ransomware triage that treats uncertainty as a first-class control variable. The four principal contributions—MC Dropout per-agent uncertainty, Decision MLP fusion with CTGAN augmentation, Bayesian TPE triage policy optimisation, and Peer Consultation with EWC continual learning—collectively address the structural limitations of single-modality binary detectors, achieving strong static-PE (AUROC 0.996) and runtime behavioural detection (AUROC 0.961), a 70% reduction in unnecessary escalation over fixed-threshold baselines, and calibration (ECE 0.018) sufficient for trustworthy autonomous containment. When the system encounters novel families or concurrent execution beyond its training distribution, epistemic uncertainty $\sigma_{\max}$ rises and the triage policy routes to Escalate—preserving analyst oversight where the model is least reliable—while Mondrian conformal prediction provides a complementary distribution-free statistical guarantee.

Future work will address process-specific telemetry to resolve the semantic gap under concurrent execution,

extension to post-2022 ransomware families (Cl0p, BlackCat/ALPHV, LockBit 3.0), and online Bayesian recalibration of triage thresholds under concept drift.

Appendix A
Algorithms and Extended Results

The behavioural feature extraction procedure (Algorithm S1), the Decision MLP training procedure with CTGAN and Bayesian TPE (Algorithm S2), per-class confusion matrices, an extended Hirano and Kobayashi [4] comparison, adaptive thresholding, and full computational complexity derivations are provided in the supplementary material.

Acknowledgements

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Code and Data Availability

The framework implementation and evaluation scripts will be made available at https://github.com/Henrinnes/Agentic_SABRE upon acceptance. The RanSMAP [4] and RDset [30] datasets are publicly available as cited. The RansKnow threat-intelligence corpus [32] will be released on Kaggle upon acceptance.

References

[1] C. Beaman, A. Barkworth, T. D. Akande, S. Hakak, and M. K. Khan, "Ransomware: Recent advances, analysis, challenges and future research directions," Comput. Secur., vol. 111, p. 102490, 2021. [Online]. Available: https://doi.org/10.1016/j.cose.2021.102490

[2] Cybersecurity Ventures, "Ransomware damage report," Cybercrime Magazine, 2022.

[3] M. Hirano and R. Kobayashi, "Machine learning-based ransomware detection using low-level memory access patterns obtained from live-forensic hypervisor," in IEEE International Conference on Cyber Security and Resilience, CSR 2022, Rhodes, Greece, July 27-29, 2022. IEEE, 2022, pp. 323–330. [Online]. Available: https://doi.org/10.1109/CSR54599.2022.9850340

[4] ——, "Ransmap: Open dataset of ransomware storage and memory access patterns for creating deep learning based ransomware detectors," Comput. Secur., vol. 150, p. 104202, 2025. [Online]. Available: https://doi.org/10.1016/j.cose.2024.104202

[5] N. Scaife, H. Carter, P. Traynor, and K. R. B. Butler, "CryptoLock (and drop it): Stopping ransomware attacks on user data," in IEEE International Conference on Distributed Computing Systems (ICDCS), 2016, pp. 303–312.

[6] D. W. Fernando and N. Komninos, "Fesad ransomware detection framework with machine learning using adaption to concept drift," Comput. Secur., vol. 137, p. 103629, 2024. [Online]. Available: https://doi.org/10.1016/j.cose.2023.103629

[7] D. Sgandurra, L. Muñoz-González, R. Mohsen, and E. C. Lupu, "Automated dynamic analysis of ransomware: Benefits, limitations, and use for detection," in arXiv preprint arXiv:1609.03020, 2016.

[8] R. Moussaileb, N. Cuppens, J. Lanet, and H. L. Bouder, "A survey on windows-based ransomware taxonomy and detection mechanisms," ACM Comput. Surv., vol. 54, no. 6, pp. 117:1–117:36, 2022. [Online]. Available: https://doi.org/10.1145/3453153

[9] J. Ispahany, M. R. Islam, M. Z. Islam, and M. A. Khan, "Ransomware detection using machine learning: A review, research limitations and future directions," IEEE Access, vol. 12, pp. 68 785–68 813, 2024. [Online]. Available: https://doi.org/10.1109/ACCESS.2024.3397921

[10] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in Proceedings of the 33nd International Conference on Machine Learning, ICML 2016, New York City, NY, USA, June 19-24, 2016, ser. JMLR Workshop and Conference Proceedings, M. Balcan and K. Q. Weinberger, Eds., vol. 48. JMLR.org, 2016, pp. 1050–1059. [Online]. Available: http://proceedings.mlr.press/v48/gal16.html

[11] J. Gawlikowski, C. R. N. Tassi, M. Ali, J. Lee, M. Humt, J. Feng, A. M. Kruspe, R. Triebel, P. Jung, R. Roscher, M. Shahzad, W. Yang, R. Bamler, and X. Zhu, "A survey of uncertainty in deep neural networks," Artif. Intell. Rev., vol. 56, no. S1, pp. 1513–1589, 2023. [Online]. Available: https://doi.org/10.1007/s10462-023-10562-9

[12] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[13] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman, "Inductive confidence machines for regression," in Machine Learning: ECML 2002. Springer, 2002, pp. 345–356.

[14] V. Vovk, I. Nouretdinov, and A. Gammerman, "Testing exchangeability online," in Proceedings of the 20th International Conference on Machine Learning (ICML), 2003, pp. 768–775.

[15] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," arXiv preprint arXiv:2208.02814, 2022.

[16] B. G. Doan, D. Q. Nguyen, P. Montague, T. Abraham, O. Y. de Vel, S. Camtepe, S. S. Kanhere, E. Abbasnejad, and D. C. Ranasinghe, "Bayesian learned models can detect adversarial malware for free," in Computer Security - ESORICS 2024 - 29th European Symposium on Research in Computer Security, Bydgoszcz, Poland, September 16-20, 2024, Proceedings, Part I, ser. Lecture Notes in Computer Science, J. García-Alfaro, R. Kozik, M. Choras, and S. K. Katsikas, Eds., vol. 14982. Springer, 2024, pp. 45–65. [Online]. Available: https://doi.org/10.1007/978-3-031-70879-4_3

[17] O. Sagi and L. Rokach, "Ensemble learning: A survey," WIREs Data Mining and Knowledge Discovery, vol. 8, no. 4, p. e1249, 2018.

[18] M. Masum, M. J. H. Faruk, H. Shahriar, K. Qian, D. Lo, and M. I. Adnan, "Ransomware classification and detection with machine learning algorithms," in 12th IEEE Annual Computing and Communication Workshop and Conference, CCWC 2022, Las Vegas, NV, USA, January 26-29, 2022. IEEE, 2022, pp. 316–322. [Online]. Available: https://doi.org/10.1109/CCWC54503.2022.9720869

[19] M. Soltani, K. Khajavi, M. J. Siavoshani, and A. H. Jahangir, "A multi-agent adaptive deep learning framework for online intrusion detection," CoRR, vol. abs/2303.02622, 2023. [Online]. Available: https://doi.org/10.48550/arXiv.2303.02622

[20] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," J. Artif. Intell. Res., vol. 16, pp. 321–357, 2002. [Online]. Available: https://doi.org/10.1613/jair.953

[21] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.

[22] U. Fiore, A. D. Santis, F. Perla, P. Zanetti, and F. Palmieri, "Using generative adversarial networks for improving classification effectiveness in credit card fraud detection," Information Sciences, vol. 479, pp. 448–455, 2019.

[23] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in Advances in Knowledge Discovery and Data Mining (PAKDD), 2022, pp. 79–91.

[24] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," Proceedings of the National Academy of Sciences, vol. 114, no. 13, pp. 3521–3526, 2017.

[25] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in USENIX Security Symposium, 2019, pp. 729–746.

[26] R. Jordaney, K. Sharad, S. K. Dash, Z. Wang, D. Papini, I. Nouretdinov, and L. Cavallaro, "Transcend: Detecting concept drift in malware classification models," in USENIX Security Symposium, 2017, pp. 625–642.

[27] F. Ceschin, M. Botacin, H. M. Gomes, F. A. Pinage, L. S. Oliveira, and A. Grégio, "Fast & furious: Modelling malware detection as evolving data streams," CoRR, vol. abs/2205.12311, 2022. [Online]. Available: https://doi.org/10.48550/arXiv.2205.12311

[28] H. Kabuye, "Agentic SABRE: An uncertainty-aware neuro-symbolic multi-agent framework for adaptive ransomware detection," Expert Systems with Applications, 2026, under review.

[29] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[30] H. Kabuye, B. Issac, R. Yumlembam, and J. Neera, "Explainable and uncertainty aware ai-based ransomware detection," IEEE Access, vol. 13, pp. 106 573–106 589, 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3581424

[31] M. Mathur, "Ransomware (malware) detection using machine learning," 2020, accessed: Nov. 9, 2023. [Online]. Available: https://github.com/muditmathur2020/RansomwareDetection/tree/master

[32] H. Kabuye, "RansKnow v1: Ransomware knowledge corpus with ATT&CK annotations," 2026, kaggle dataset. Accessed June 2026. [Online]. Available: https://www.kaggle.com/datasets/henrykabuye/ransknow-v1

[33] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in Proceedings of the 34th International Conference on Machine Learning (ICML), 2017, pp. 1321–1330.

[34] J. S. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in Advances in Neural Information Processing Systems (NeurIPS), vol. 24, 2011.